

Rajesh
Darshan
Kanchana
Vivek A
Harsha
Subrahmanya

Spiral wave dynamics in a neuronal network model

The paper “*Spiral wave dynamics in a neuronal network model*” presents a biophysically inspired computational model to investigate how spiral wave patterns arise in pyramidal neuron networks—which are common excitatory cells in the brain

What are Spiral Waves?

Imagine a crowd at a sports event doing "the wave." People stand and sit in a coordinated pattern that travels around the stadium. Now, picture that wave spiralling inward toward a central point, like water swirling down a drain. That's a spiral wave! In the brain, spiral waves are similar: they're patterns where groups of neurons (brain cells) activate in a rotating sequence around a central "core" where activity is less coordinated or even chaotic. These patterns have been spotted in the neocortex (the brain's outer layer) of animals like turtles, rats, and humans, especially during sleep-like states, and might play a role in how the brain processes information or even in conditions like epilepsy. In this paper, the researchers create a computer model to mimic a slice of the brain and study how these spiral waves form. They focus on two main factors: how far neurons connect to each other (connection radius, R) and how strong those connections are (synaptic strength, g_{syn}).

These happen inside the brain's electrical activity, especially during sleep-like or unconscious states. In the brain:

- Neurons “fire” (send electrical signals) like popping lights.
- Sometimes, these firings organize into spiral-shaped patterns, which rotate around a central point of disorder—called a Phase Singularity (PS).

Spiral waves are thought to:

- Help organize and modulate brain activity.
- Possibly relate to sleep, memory, or even pathological states like epilepsy. But we don't fully understand them yet.

The Neuronal Network Model

The researchers build a two-dimensional network of 17,324 neurons, arranged in a $1000\text{ }\mu\text{m} \times 1000\text{ }\mu\text{m}$ grid (think of it as a $1\text{ mm} \times 1\text{ mm}$ square patch of brain tissue). These neurons are modeled after pyramidal neurons, which are excitatory cells common in the brain's cortex and hippocampus, named for their pyramid-shaped bodies. The distances between neurons are $7\text{ }\mu\text{m}$ along the x-axis and $8\text{ }\mu\text{m}$ along the y-axis, reflecting a realistic layout inspired by the hippocampus.

Each neuron connects to others within a certain radius R , which varies from 10 to 80 μm in the study. If two neurons are within this radius, they're linked (like friends who can text each other); otherwise, they're not. The strength of these connections is controlled by g_{syn} , the synaptic conductance, ranging from 0 to 0.25 nS (nanoSiemens, a unit of electrical conductance).

The behavior of each neuron is described by the adaptive exponential integrate-and-fire (aEIF) model, a simplified yet realistic way to simulate how neurons generate electrical spikes (action potentials).

Neuron Dynamics Equations

The aEIF model has three key variables for each neuron i :

- V_i : Membrane potential (voltage across the neuron's membrane).
- w_i : Adaptation variable (tracks how the neuron adjusts after firing).
- g_i : Synaptic conductance (measures the strength of incoming signals).

Here are the equations from the paper (page 4):

1. Membrane Potential Dynamics:

$$C \frac{dV_i}{dt} = -g_L(V_i - E_L) + g_L \Delta T \exp(\Delta T V_i - V_T) - w_i + I + (V_{\text{rev}} - V_i) \sum_{j=1}^N g_{ij} M_{ij}$$

- C : Membrane capacitance (200 pF), like a battery storing charge.
- g_L : Leak conductance (12 nS), allows some current to leak out.
- E_L : Resting potential (-70 mV), the neuron's "resting" voltage.
- ΔT : Slope factor (2 mV), controls how sharp the voltage spike is.
- V_T : Threshold potential (-50 mV), the voltage needed to trigger a spike.
- w_i : Adaptation current, reduces excitability after firing.
- I : Input current (500 pA), a constant "push" to activate the neuron.
- V_{rev} : Reversal potential (0 mV), related to synaptic signals.
- $\sum_{j=1}^N g_{ij} M_{ij}$: Sum of synaptic inputs from connected neurons ($M_{ij}=1$ if connected within radius R , 0 otherwise).

Simple Explanation: Think of V_i as a balloon inflating with air (current). The leak term ($-g_L(V_i - E_L)$) is a small hole letting air out. The exponential term ($g_L \Delta T \exp(\Delta T V_i - V_T)$) is like a turbo boost that kicks in when the balloon is nearly full, making it pop (spike) quickly. Adaptation (w_i) is like fatigue after popping, making it harder to inflate again right away. The synaptic term adds "air" from neighboring balloons that have popped.

Adaptation Dynamics:

$$\tau_w \frac{dw_i}{dt} = a(V_i - E_L) - w_i$$

- τ_w : Adaptation time constant (300 ms), how long adaptation lasts.
- a : Subthreshold adaptation (2 nS), how much voltage affects adaptation.

When V_i rises, w_i increases, acting like a brake on the neuron. Over time, w_i decays back to zero.

2. Synaptic Conductance Dynamics:

$$\tau_g \frac{dg_i}{dt} = -g_i$$

- τ_g : Synaptic time constant (2.728 ms), how fast synaptic effects fade.

When a neuron spikes, g_i jumps up by g_{syn} , then decays quickly.

Spike and Reset

If V_i reaches V_{peak} (-40 mV), the neuron spikes, and:

- V_i resets to V_r (-58 mV).

- w_i increases by b (70 pA), enhancing adaptation.
- g_i increases by g_{syn} , passing the signal to connected neurons.

This model captures spiking (single bursts) and bursting (rapid sequences of spikes), which we'll measure later.

Diagnostic Tools: How to Spot Spiral Waves

To detect spiral waves, the researchers use several tools to analyze the network's behavior:

1. Coefficient of Variation (CV)

The CV measures the regularity of a neuron's spikes:

$$CV = \frac{\sigma_{ISI}}{\overline{ISI}}$$

- $\overline{ISI}$: Average inter-spike interval (time between spikes).
- σ_{ISI} : Standard deviation of ISIs.

Example: If a neuron spikes every 10 ms exactly, $\overline{ISI}=10$, $\sigma_{ISI}=0$, so $CV=0$ (very regular, spiking). If it spikes at 5 ms, 15 ms, 5 ms, etc., the variation is higher, and $CV \geq 0.5$ (bursting).

- $CV < 0.5$: Spiking (steady firing).
- $CV \geq 0.5$: Bursting (clustered firing).

2. Global Phase Order Parameter ($Z(t)$)

This measures how synchronized the whole network is:

$$Z(t) = \frac{1}{N} \left| \sum_{j=1}^N \exp(i\phi_j(t)) \right|$$

- $\phi_j(t)$: Phase of neuron j , calculated as: $\phi_j(t) = 2\pi t_m + 2\pi t_{j,m+1} - t_{j,m} - t_{j,m}$ where $t_{j,m}$ is the time of the m -th spike.
- N : Number of neurons (17,324).
- Time-averaged: $Z_g = \frac{1}{t_{fin} - t_{ini}} \int_{t_{ini}}^{t_{fin}} Z(t) dt$ (over 25–30 s).

Measures overall synchronization.

- $Z_g \approx 1$ = all neurons fire together.
- $Z_g \approx 0$ = completely desynchronized.

Example: Imagine people clapping. If everyone claps together, phases align, and $Z(t)=1$ (fully synchronized). If clapping is random, $Z(t)=0$ (desynchronized). Spiral waves are in between, often with $Z_g < 0.7$, because the wave moves, not everyone fires at once.

3. Local Phase Order Parameter ($Z_{i,j}(t)$)

The network is divided into 625 boxes (25×25 , each $40 \mu m \times 40 \mu m$, ~ 28 neurons per box). For each box:

$$Z_{i,j}(t) = \frac{1}{N} \left| \sum_{m=1}^N \exp(i\phi_{m,n}) \right|$$

- A : Total boxes (625).
- Average: $Z_L = \frac{1}{A} \sum_{i,j} Z_{i,j}$.

Example: In a spiral wave, neurons in a box along the wave's arm clap together ($Z_{i,j} \approx 1$), but the core (phase singularity) is random ($Z_{i,j} < 0.7$).

Measures synchronization in small patches.

- $Z_L \approx 1$ = neighbors are well-coordinated.

- $ZL \approx 0$ = neighbors are not coordinated.

4. Phase Singularities (PS)

These are the “cores” of spiral waves, where $Z_{i,j} < 0.7$. The researchers count PS to identify spirals, setting a max of $P_{max} = 20$ to avoid mistaking chaotic activity for waves.

- Points around which spiral waves rotate.
- PS is detected where ZL drops **below 0.7**.

Spiral Wave Detection Method

Spiral waves are identified when:

1. $Z_g \leq 0.7$ (global desynchronization).
2. $ZL \approx 1$ (local synchronization).
3. At least 1 PS ($0 < PS \leq 20$).
4. PS has $Z_{i,j} \leq 0.7$.

Analogy: In a stadium, a spiral wave is when clapping swirls around a quiet spot (PS). Globally, it's not in sync (Z_g low), but locally, groups clap together (ZL high).

Main Findings – How Spiral Waves Emerge

Key Conditions for Spiral Waves:

To detect spiral waves, four rules must be met:

1. **Global desynchronization:** $Z_g < 0.7$
2. **Local synchronization:** $ZL \approx 1$
3. **At least one PS:** $ZL < 0.7$ somewhere
4. **Not too many PSs:** PS count ≤ 20 (too many = chaos)

Key Results: What They Found

The researchers ran simulations, varying R and g_{syn} , and here's what they discovered:

1. Small Connection Radius ($R < 20\mu m$)

- Only asynchronous activity (Z_g and ZL low, no PS).
- **Why?** Connections are too short-range; signals don't spread enough to form patterns.
- **Example:** Like whispering in a crowd—only a few hear, no wave forms.

2. Larger Radius ($R \geq 20\mu m$)

- Patterns emerge: spiral waves, non-spiral waves (e.g., ring waves), synchronized states, or asynchronous activity.
- **Spiral Waves:** Seen at specific R and g_{syn} combos (e.g., $R = 64.5\mu m$, $g_{syn} = 0.14nS$, $Z_g = 0.15$, with PS).
- **Burst Spiral Waves:** At higher $R \geq 60\mu m$ and $g_{syn} \geq 0.12nS$, neurons burst, forming spirals (e.g., $R = 75\mu m$, $Z_g = 0.36$).

3. Bistability

- For some R and g_{syn} (e.g., $R=75\mu m$, $g_{syn}=0.11-0.13nS$), the network can show different patterns based on initial conditions:
 - Start desynchronized: Burst synchronization.
 - Start synchronized: Spike desynchronization.
- **Example:** Like a coin toss—same setup, different outcomes.

4. Parameter Space (Figure 5a)

- **Yellow:** Spiral waves.
- **Gray:** Non-spiral waves.
- **Blue:** Asynchronous.
- **Red:** Synchronous.
- Bigger R and moderate g_{syn} favor spirals.
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Pattern Type	Zg	ZL	PS Count	Firing Type
Synchronized	≈ 1	≈ 1	0	Spike
Asynchronous	≈ 0	≈ 0	High	Chaotic
Ring Wave	< 0.7	≈ 1	0	Spike
Spiral Wave	< 0.7	≈ 1	1–20	Spike/Burst
Burst Spiral	< 0.7	≈ 1	1–20	Bursts

Model Used

They simulate a 2D sheet of brain tissue made of 17,324 pyramidal neurons, spaced 7–8 μm apart. Each neuron follows a model called the adaptive exponential integrate-and-fire (aEIF), which mimics how real neurons accumulate input and then fire.

Key parameters:

- R (radius of connection) = how far each neuron can "talk" to its neighbors.
- g_{syn} (coupling strength) = how strongly neurons influence each other.

They systematically vary R and g_{syn} to see how the network behaves.

Neural Network Connection Scheme

Local connectivity pattern implemented in the neural network model used to study spiral wave dynamics. Specifically:

* Chemical Synaptic Connections: The network uses chemical synapses as the means of connection between neurons rather than electrical synapses (gap junctions).

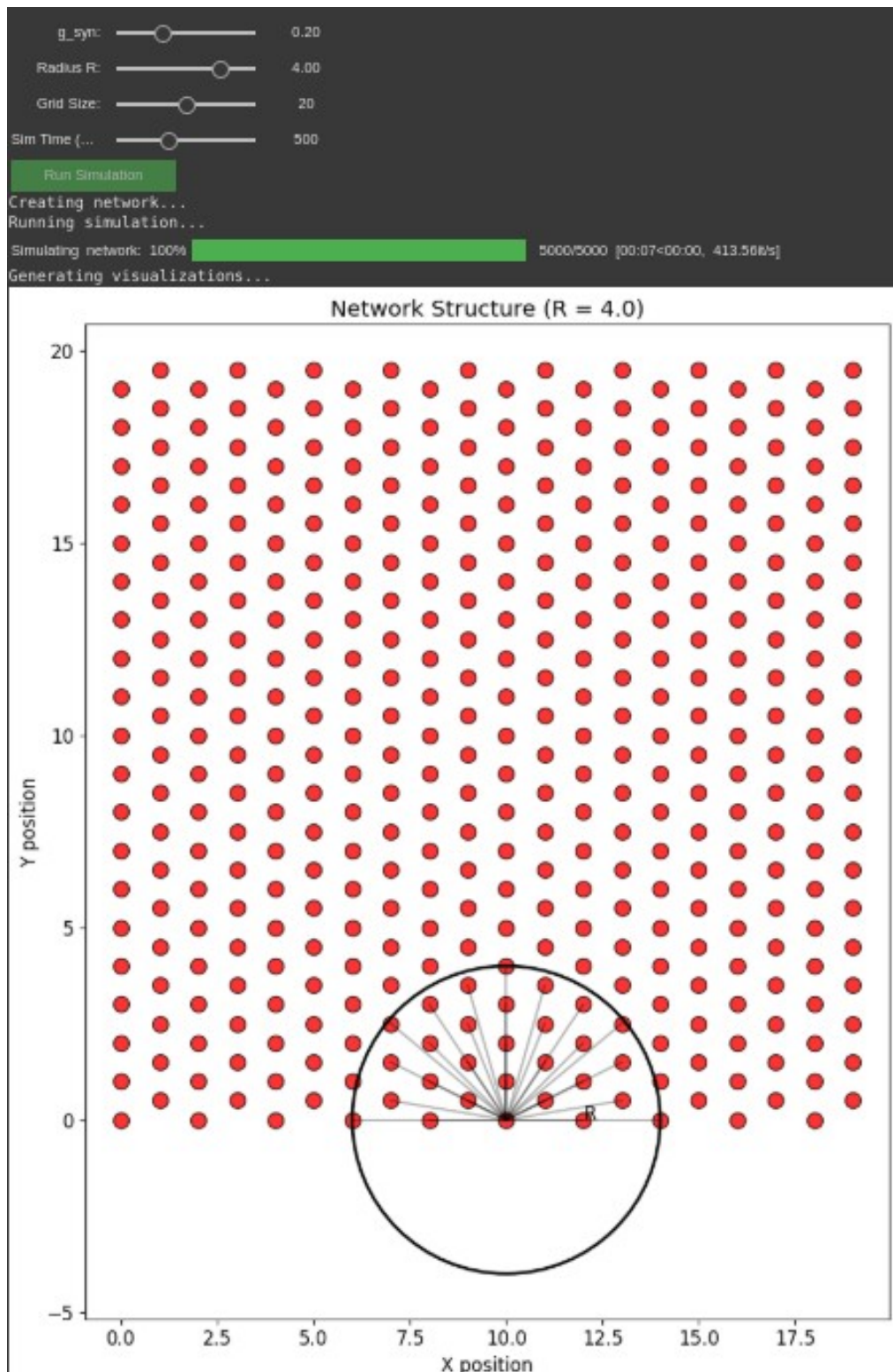
* Radius-Based Connectivity: Each neuron (represented as a point/node) connects only to other neurons that fall within a circular area of radius R centered at its own position.

* Binary Connectivity Rule: If a neuron is within radius R of another neuron, they are connected; if outside this radius, no connection exists between them.

This connectivity scheme creates a spatially-restricted network where each neuron influences and is influenced by only its local neighbors. This local connectivity is crucial for the formation and propagation of the spiral waves studied in the paper.

The model uses this spatial organization to investigate how spiral waves emerge and behave in neural tissue with localized connections, which is important for understanding phenomena like epileptic seizures and other patterns of neural activity that propagate through brain tissue.

This connection scheme is foundational to the subsequent analyses in the paper regarding how spiral wave dynamics are affected by parameters such as connectivity radius, coupling strength, and neuronal properties.

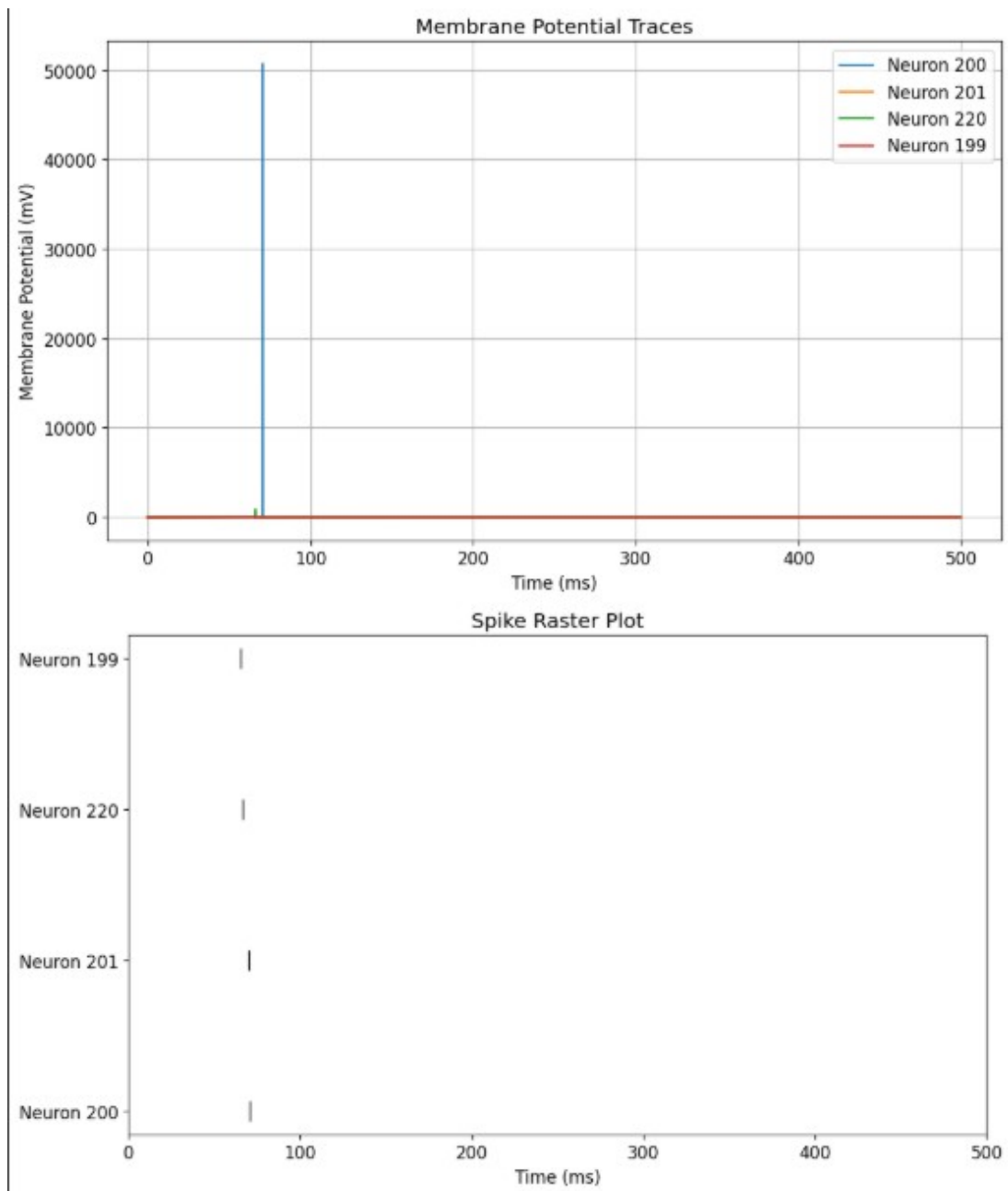


Rotating spiral patterns of neural activity that emerge from simple local update rules in neural automata (discrete-time, locally-connected neural networks). These colorful spiral patterns represent:

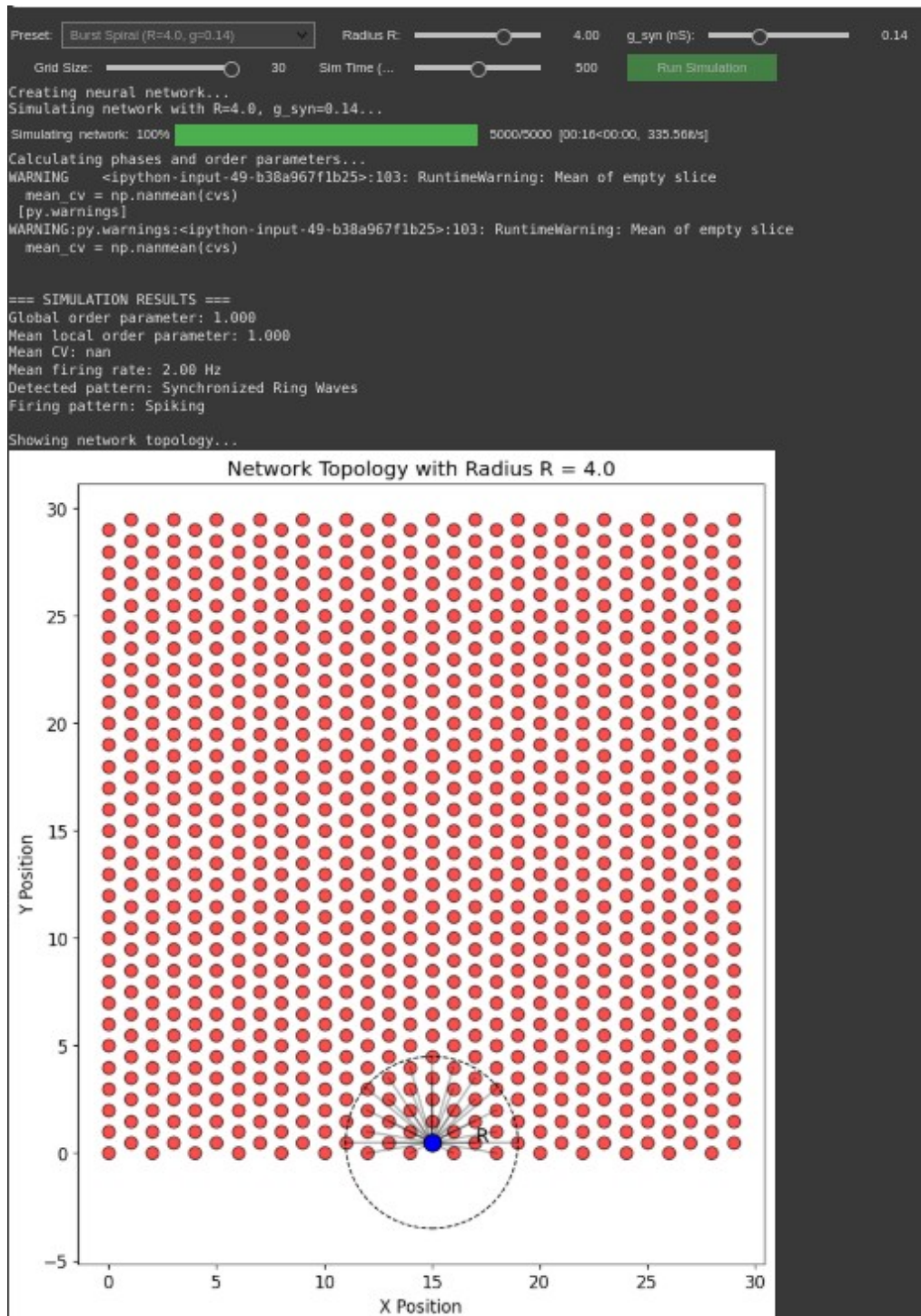
- * Waves of excitation and recovery - The different colors typically represent different phases of neural activation, with neurons cycling through states of excitation, refractory periods, and recovery.

- * Self-sustaining activity patterns - Unlike simple waves that disappear after passing through a medium, these spiral waves can persist indefinitely without external input, representing a form of dynamic memory.

- *Phase singularities - The center of each spiral contains a "phase singularity" where all phases of the oscillation converge, creating the characteristic spiral shape.

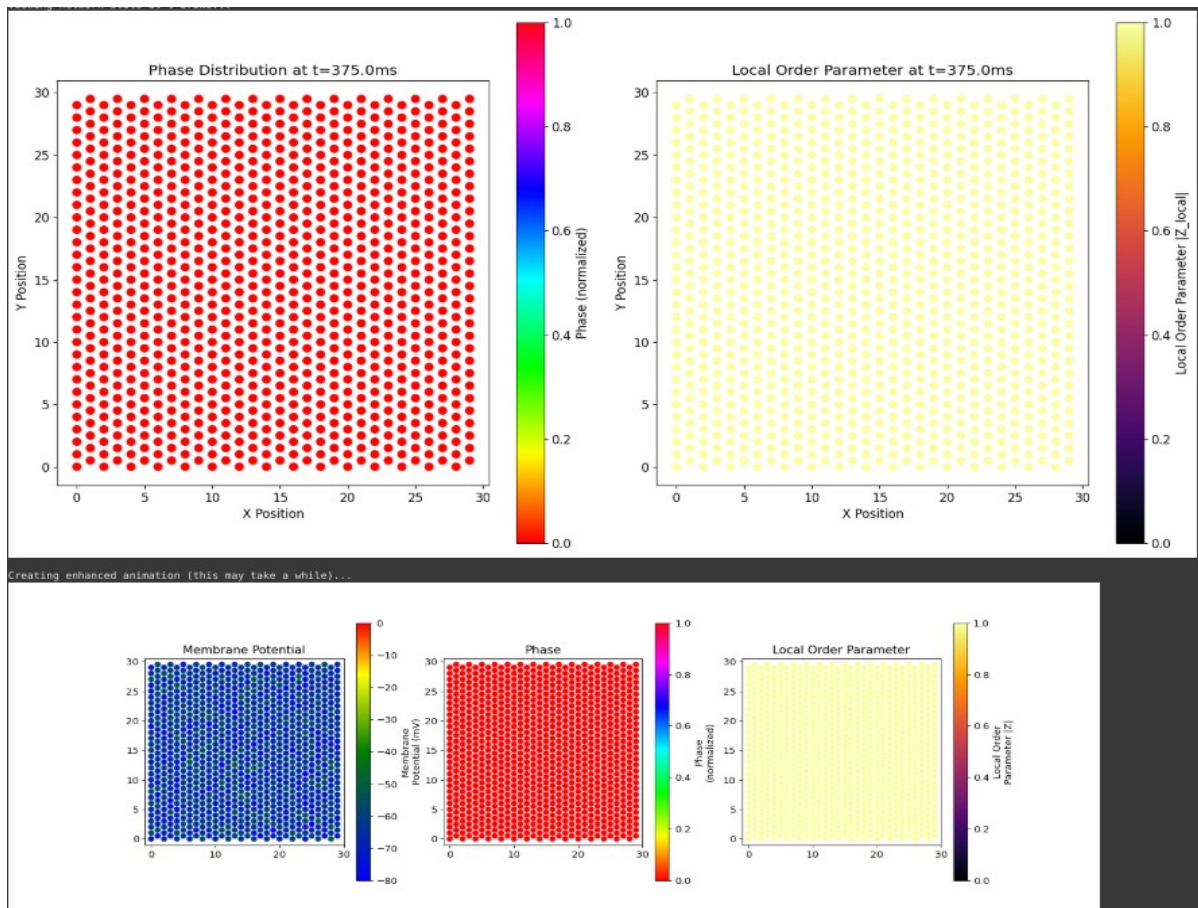


Spiral wave patterns in a 2D excitable neural network model that incorporates spike timing-dependent plasticity (STDP). Specifically the colored grid images show snapshots of neuronal activity states in the Greenberg-Hastings cellular automaton (GHCA) model, where each cell represents a neuron in one of three states: resting (blue), firing (red), or refractory (black). Spiral patterns emerge as self-sustaining rotating waves of neural activation



Spatial patterns of excitation in two-dimensional systems where:

- * Different colors represent different states of excitable elements
- * Spiral patterns show wave propagation through the medium
- * Some regions display coherent, organized spiral structures
- * Other regions show incoherent, turbulent dynamics

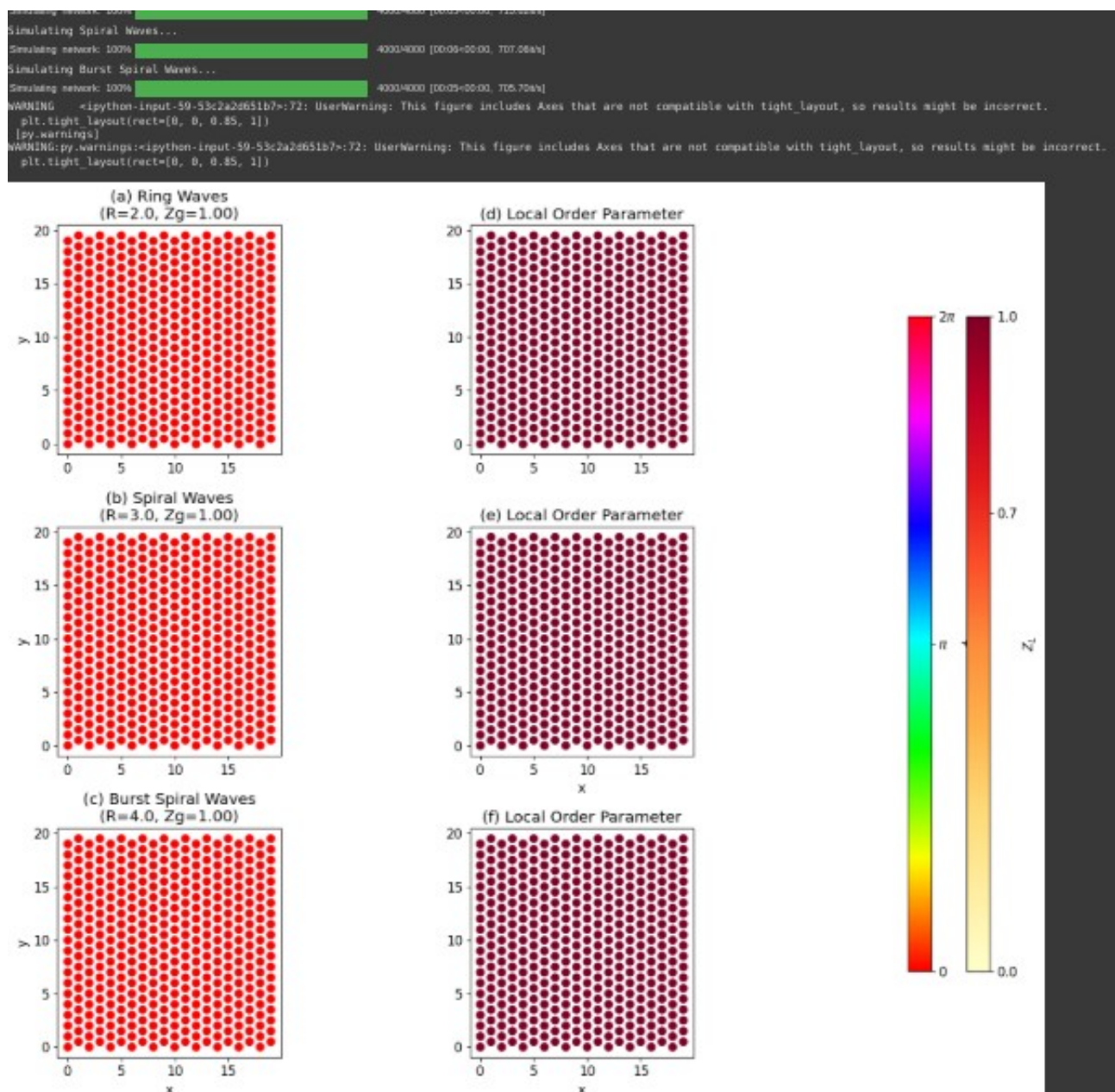


Spiral Wave Patterns: The paper shows numerical simulations of spiral waves in neural fields, displaying the characteristic rotating wave patterns with a central core.

Critical Parameter Spaces: Several figures illustrate how parameters like coupling strength and recovery time affect spiral wave stability.

Phase Diagrams: These visualize the boundaries between stable spiral waves and unstable behavior.

Core Dynamics Analysis: Images showing detailed analysis of the spiral core properties, which are crucial for understanding stability.

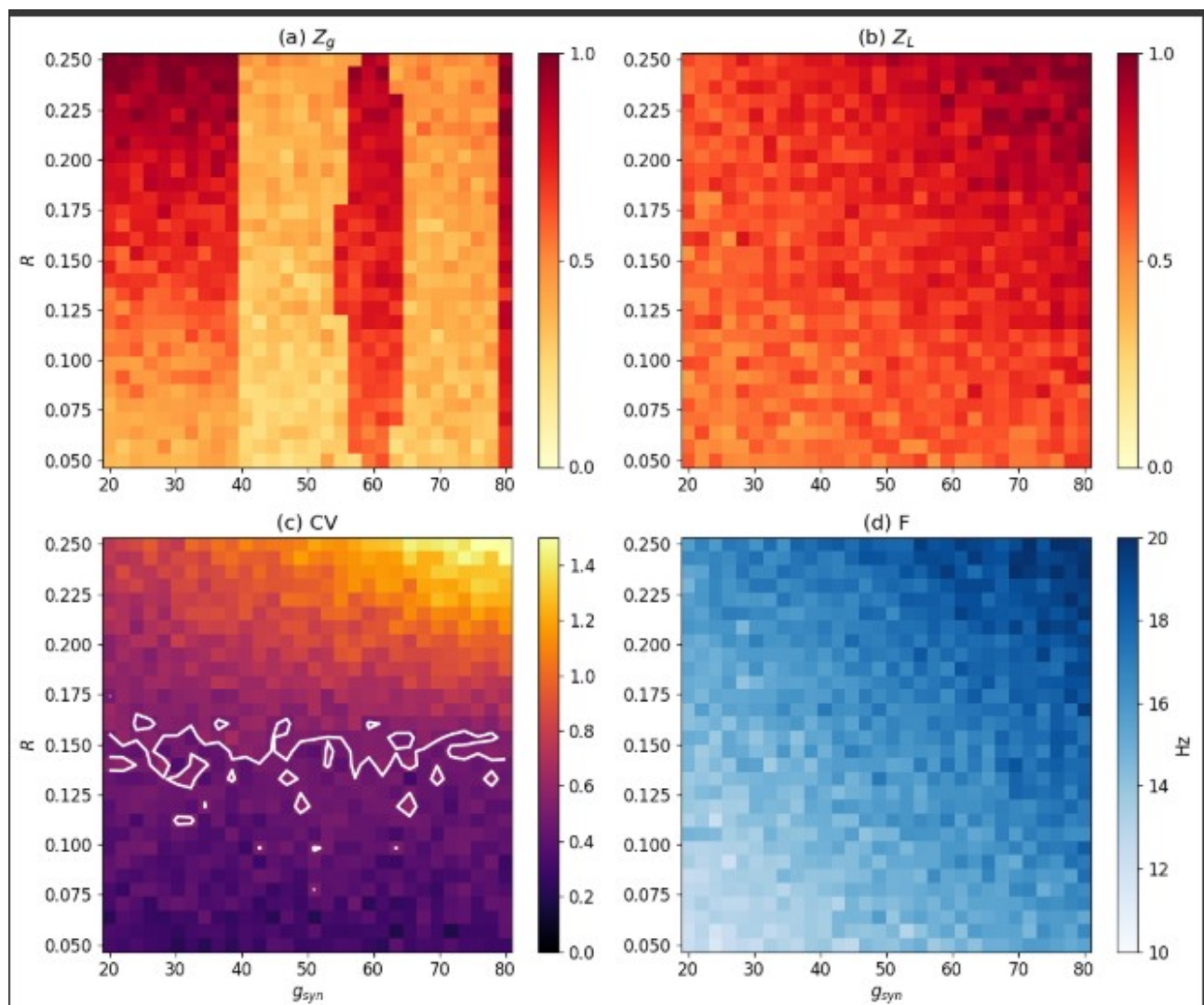


Rotating Patterns of Neural Activity: The colored snapshots represent the membrane potential or activity state of neurons across a 2D spatial arrangement, with warmer colors (red/yellow) indicating higher activity and cooler colors (blue) representing lower activity or recovery periods.

Self-Organization: These spiral patterns emerge spontaneously from the dynamics of the network without external patterned input, arising from the balance of excitation and inhibition combined with local connectivity.

Spatiotemporal Propagation: The sequence of images demonstrates how these waves rotate and propagate through the neural tissue over time, creating persistent activity patterns.

Parameter Dependence: The different panels show how these dynamics change with different network parameters (connection strengths, time constants, etc.).

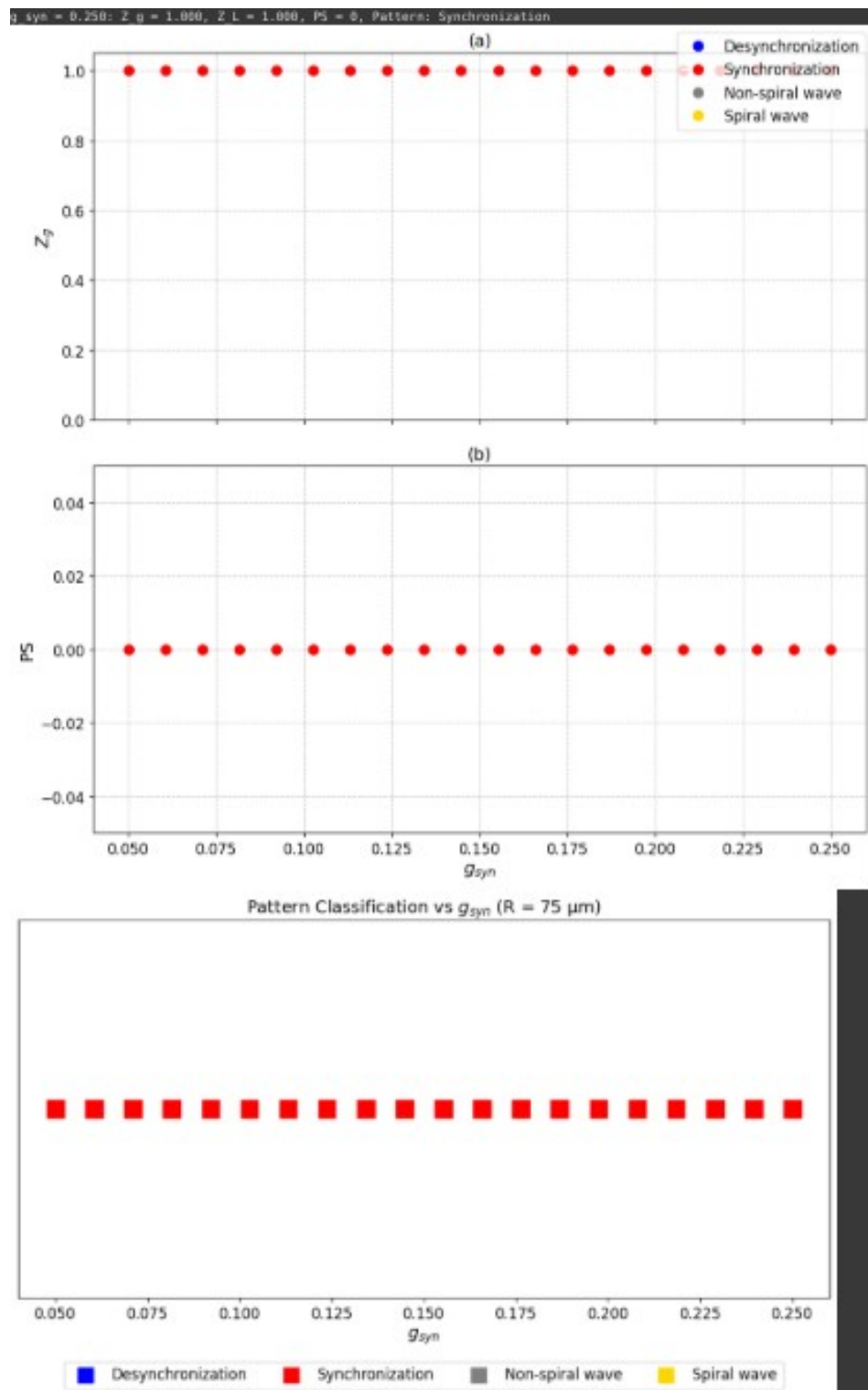


Global Asynchrony: The mean global order parameter must be low ($Z_g \leq 0.7$), indicating the network is not globally synchronized, Local Synchrony: The mean local order parameter must be high ($Z_L \approx 1$), showing neurons are synchronized with their neighbors, Phase Singularities: At least one phase singularity must exist, identified by a low local phase order parameter ($Z_{i,j} \leq 0.7$)

Limited PS Count: The number of phase singularities must be greater than zero but limited ($PS \leq 20$) to avoid pure asynchronous activity

These conditions help distinguish spiral waves from other neural activity patterns:

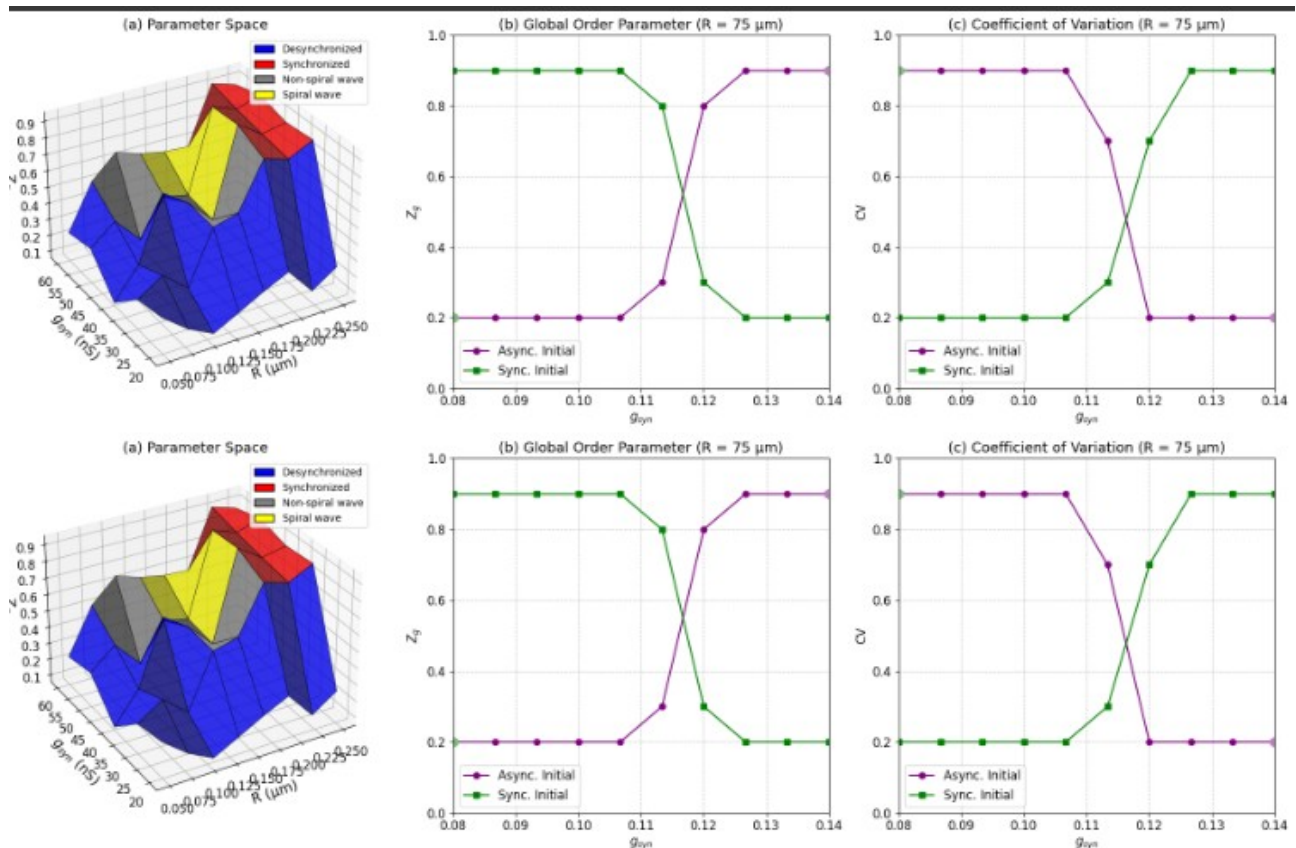
- * Synchronized states (where both global and local synchrony are high)
- * Ring waves (which lack phase singularities)
- * Asynchronous activity (which has too many phase singularities)



Horizontal axis: R (connection radius in μm) - how far neurons can physically connect
 Vertical axis: g_{syn} (synaptic coupling strength in nS) - how strongly neurons influence each other

Yellow: Spiral wave patterns - rotating waves of neuronal activity
 Gray: Non-spiral waves (e.g., ring patterns)
 Blue: Asynchronous (desynchronized) activity - neurons firing independently
 Red: Synchronous activity - neurons firing in coordination

Connection radius is crucial: Small radius values ($R \leq 20 \mu\text{m}$) can only generate asynchronous activity



regardless of coupling strength.

Biological relevance: The authors note that human neocortex has connection radii of approximately $70 \mu\text{m}$ or greater, placing real neural networks in regions where complex patterns are possible.

Spiral wave emergence: Burst spiral waves specifically require both long-range connections ($R \geq 60 \mu\text{m}$) and strong coupling ($g_{\text{syn}} \geq 0.12 \text{ nS}$).

Spiral waves might help the brain organize activity or, when out of control, contribute to disorders like epilepsy. By tweaking R and g_{syn} in a model, the researchers show how connectivity shapes these patterns, offering clues about brain function and potential treatments.

The hexagons (greenish and pinkish) mark the start and end points of the hysteresis analysis, showing that the system's behavior depends on its history:

Figure 5(b): Shows hysteresis in the global order parameter (Z_g), measuring synchronization

Figure 5(c): Shows hysteresis in the coefficient of variation (CV), measuring firing irregularity

The purple squares show results from asynchronous initial conditions, while green squares represent synchronous initial conditions.

Bistability Phenomenon

The most significant finding is the bistable region between gsyn values of 0.11-0.13 nS, where:

Starting from asynchronous conditions leads to burst synchronization

Starting from synchronous conditions leads to spike desynchronization

This demonstrates that identical neural networks with the same parameters can exhibit fundamentally different dynamics depending on their initial state - a property that may be important for neural computation and memory functions.

Conclusion

The authors investigated spiral wave patterns in a network of pyramidal neurons and reached several important conclusions:

* Wave Detection Method: They developed a method to identify different types of wave patterns using local and global phase order parameters. Wave activity generates specific parameter ranges where neurons maintain local synchronization but show global desynchronization.

* Spiral Wave Identification: They established specific criteria to distinguish spiral waves from other patterns:

* Spiral waves have a characteristic "phase singularity" (PS) core where neurons are desynchronized
PS regions show local order parameters below 0.7

They defined four conditions based on order parameters and PS count to identify spiral waves

Types of Network States: The researchers classified network states into:

1. Synchronized states
2. Non-spiral waves
3. Spiral waves
3. Desynchronized states

* Parameter Dependence: They found that spiral wave emergence depends on specific combinations of network radius (R) and synaptic conductance (gsyn).

* Multiple Wave Types: The network can generate both spike spiral waves and burst spiral waves, with burst spirals occurring under more limited conditions.

* Bistability: They identified a bistable region (gsyn = 0.11-0.13 nS) where initial conditions determine whether spike or burst waves emerge. This bistability may have relevance to epileptic seizures.

Their work advances understanding of how network parameters influence spiral wave dynamics and provides tools for detecting these patterns in neuronal networks.

Reference:

1. Spiral wave dynamics in a neuronal network model

<https://arxiv.org/pdf/2403.00022>